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i. e., stationary state changes in time only by a phase factor (**NO perturbation**)

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II. time evolution of a **coupled two-level system** - time-independ. perturbation

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$$\therefore \mathrm{U}(t, t_0) \overset{t_0=0}{=} \begin{pmatrix} \cos\frac{\Omega t}{2} - i\cos 2\theta \sin\frac{\Omega t}{2} & -ie^{-i\chi}\sin 2\theta \sin\frac{\Omega t}{2} \\ -ie^{i\chi}\sin 2\theta \sin\frac{\Omega t}{2} & \cos\frac{\Omega t}{2} + i\cos 2\theta \sin\frac{\Omega t}{2} \end{pmatrix}; \quad \Omega = \frac{\Delta}{\hbar}$$

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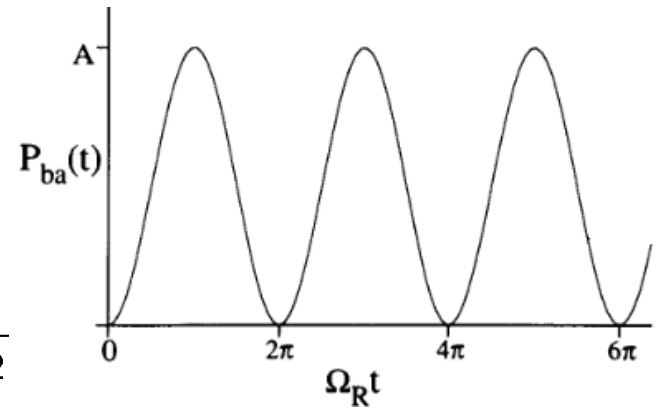
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Rabi frequency, $\Omega_R = \frac{1}{\hbar}\sqrt{4V^2+(E_a-E_b)^2}$

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TDSE : $i\hbar \sum_n \dot{c}_n(t)\psi_n = \widehat{H} \sum_m c_m(t)\psi_m$

take inner product on both sides with ψ_n, $\quad i\hbar\frac{\partial c_n}{\partial t} = \sum_m \langle\psi_n|\widehat{H}(t)|\psi_m\rangle\, c_m(t)$ .....(A)

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Multiplying through by $-(i/\hbar)e^{iE_a t/\hbar}$, $\dot{c}_a = -\frac{i}{\hbar}[c_a H'_{aa} + c_b H'_{ab}e^{-i(E_b - E_a)t/\hbar}]$

Similarly, inner product with $\psi_b \implies \dot{c}_b = -\frac{i}{\hbar}[c_b H'_{bb} + c_a H'_{ba} e^{i(E_b - E_a)t/\hbar}]$

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Typically, diagonal elements of H$'$ vanish, $H'_{aa} = H'_{bb} = 0$

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or,

$$i\dot{c}_+ = \omega_0 c_+ + \omega_1 e^{-i\omega t}c_- \quad \ldots\ldots\ldots(1)$$

$$i\dot{c}_- = \omega_1 e^{i\omega t}c_+ - \omega_0 c_- \quad \ldots\ldots\ldots(2)$$

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Substituting, get characteristic equation $\lambda^2 + \omega\lambda - \left(\omega_1^2 + \omega_0^2 - \omega\omega_0\right) = 0$
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Define $\Omega = \sqrt{(\omega_0 - \frac{1}{2}\omega)^2 + \omega_1^2} \implies \lambda_\pm = -\frac{\omega}{2} \pm \Omega$
The general solution is: $c_+(t) = ae^{i\lambda_+ t} + be^{i\lambda_- t}$
choose initial conditions, $c_+(0) = 1;\ c_-(0) = 0$
obtain: $c_+(t) = \left[\cos(\Omega t) - i\frac{\omega_0 - \frac{1}{2}\omega}{\Omega}\sin(\Omega t)\right] e^{-\frac{i\omega t}{2}}$

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Notice: $|c^-(t)|^2 = \frac{\omega_1^2}{\Omega^2}\sin^2(\Omega t)$; For $\omega = 2\omega_0$, there is a resonance

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Suppose the particle starts out in the lower state, $c_a(0) = 1, \quad c_b(0) = 0$

If there was no perturbation, they would stay this way forever:

Zeroth Order: $c_a^{(0)}(t) = 1, \quad c_b^{(0)}(t) = 0$

First Order: $\dot{c}_a = -\frac{i}{\hbar} H'_{ab} e^{-i\omega_0 t} c_b, \quad \dot{c}_b = -\frac{i}{\hbar} H'_{ba} e^{i\omega_0 t} c_a$

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insert these expressions to obtain the second-order approximation

Second Order: $\frac{dc_a^{(2)}}{dt} = -\frac{i}{\hbar} H'_{ab} e^{-i\omega_0 t} \left(-\frac{i}{\hbar}\right) \int_0^t H'_{ba}(t') e^{i\omega_0 t'} dt' \implies$

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while c_b is unchanged $(c_b^{(2)}(t) = c_b^{(1)}(t))$

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we could continue this indefinitely, inserting the nth-order approximation and solving for the $(n+1)$th order

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zeroth order contains no factors of $\hat{H}'$
first-order correction contains one factor of $\hat{H}'$

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zeroth order contains no factors of $\hat{H}'$
first-order correction contains one factor of $\hat{H}'$
second-order correction has two factors of $\hat{H}'$, and so on

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reading from right to left :

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picks up phase factor $e^{-iE_a t'/\hbar}$

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reading from right to left :

system remains in state a from time 0 to time t'

picks up phase factor $e^{-iE_a t'/\hbar}$

makes a transition from state a to state b at time t'

pictorial interpretation :

Write : $c_b^{(1)}(t)e^{-iE_b t/\hbar} = -\frac{i}{\hbar}\int_0^t e^{-iE_b(t-t')/\hbar}H'_{ba}(t')e^{-iE_a t'/\hbar}dt'$

reading from right to left :

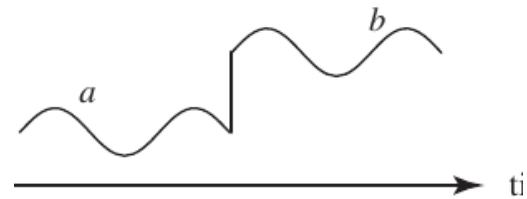
system remains in state a from time 0 to time t'

picks up phase factor $e^{-iE_a t'/\hbar}$

makes a transition from state a to state b at time t'

and then remains in state b until time t

picking up the phase factor $e^{-iE_b(t-t')/\hbar}$



$$c_a^{(2)}(t)e^{-iE_a t/\hbar} =$$
$$e^{-iE_a t/\hbar} + \left(-\frac{i}{\hbar}\right)^2 \int_0^t \int_0^{t'} e^{-iE_a(t-t')/\hbar} H'_{ab}(t') e^{-iE_b(t'-t'')/\hbar} H'_{ba}(t'') e^{-iE_a t''/\hbar} dt'' dt'$$



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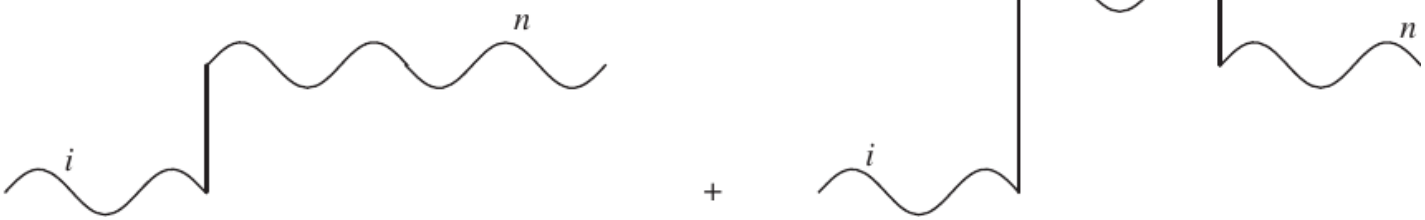
1st term : system remains in state a for the entire time

2nd term : system transitions from a to b at time t'' and then back to a at time t'

More generally, higher orders and for multi-level systems

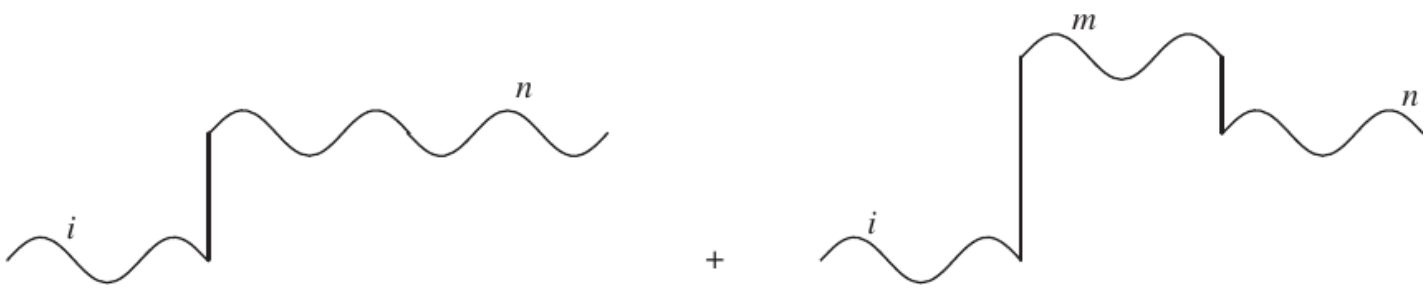
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$$c_n^{(2)}(t)e^{-iE_n t/\hbar} = \delta_{ni}e^{-iE_i t/\hbar} + \left(\frac{-i}{\hbar}\right)\int_0^t e^{-iE_n(t-t')/\hbar}H_{ni}\prime(t')e^{-iE_i t'/\hbar}dt' \quad + \quad \sum_m \left(-\frac{i}{\hbar}\right)^2 \int_0^t \int_0^{t'} e^{-iE_n(t-t')/\hbar}H'_{nm}(t')e^{-iE_m(t'-t'')/\hbar}H'_{mi}(t'')e^{-iE_i t''/\hbar}dt''dt'$$



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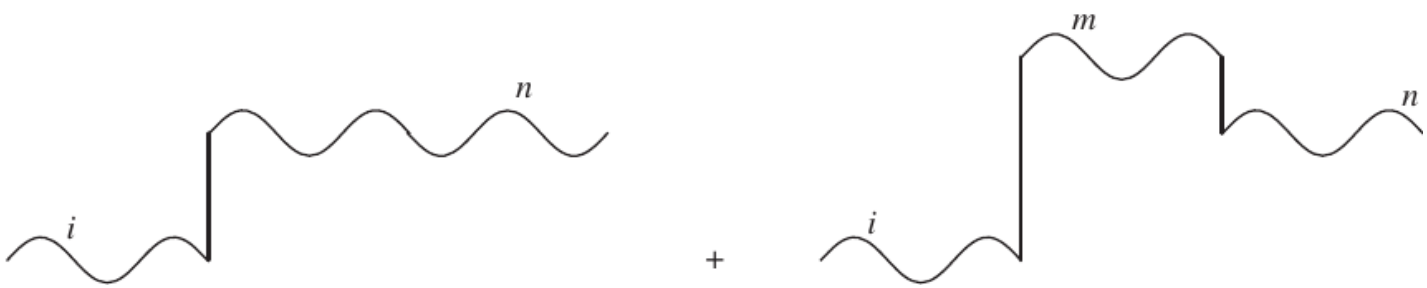
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2nd-order term : transition occurs via an intermediate (or virtual) state m

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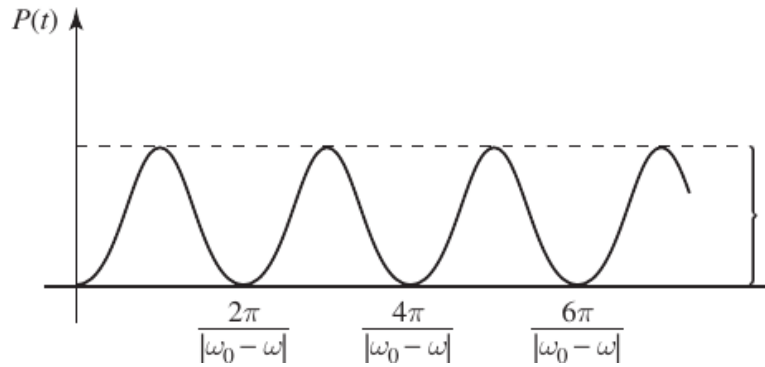
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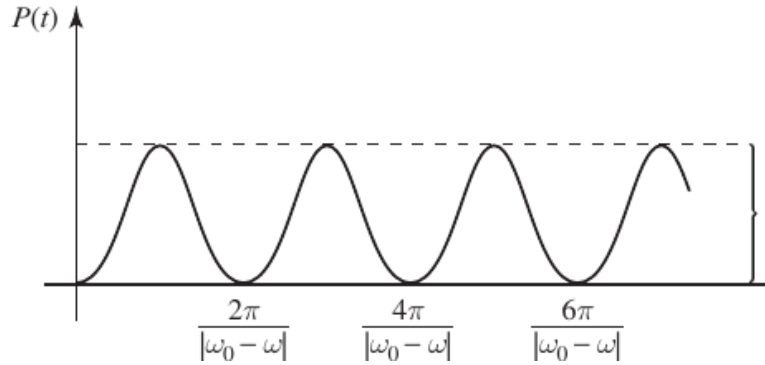
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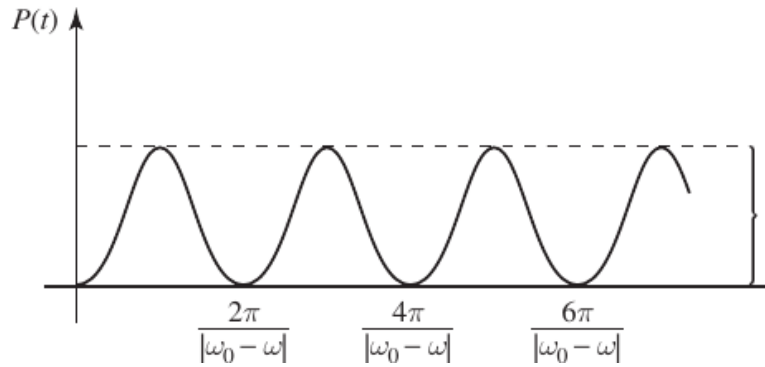
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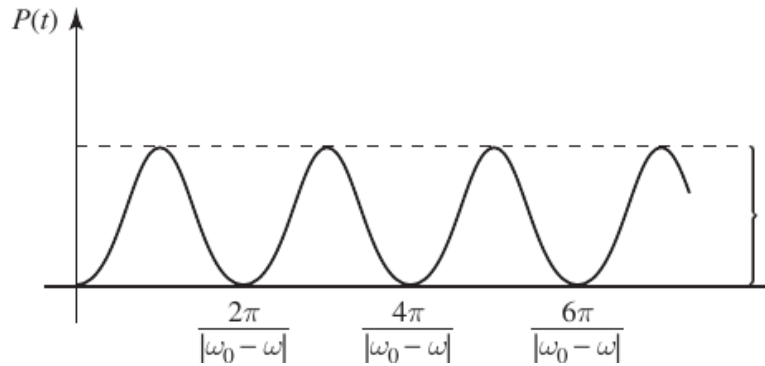
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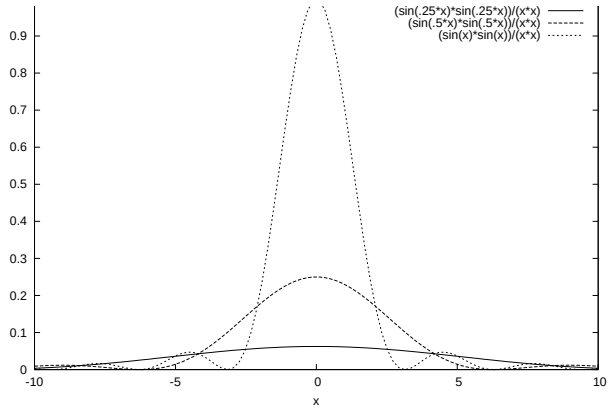


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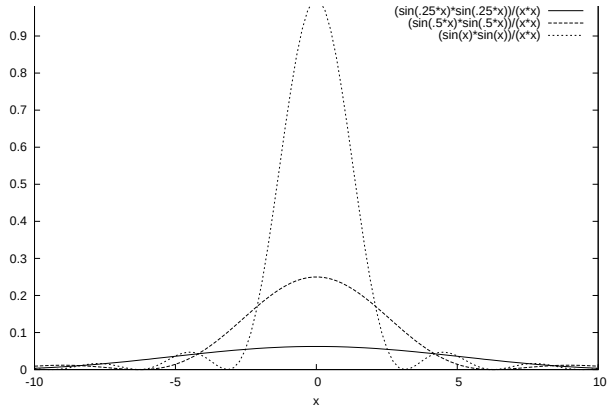
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we get Rabi oscillations

plot of $P_{a\to b}$ with ω changes with t and with increasing time

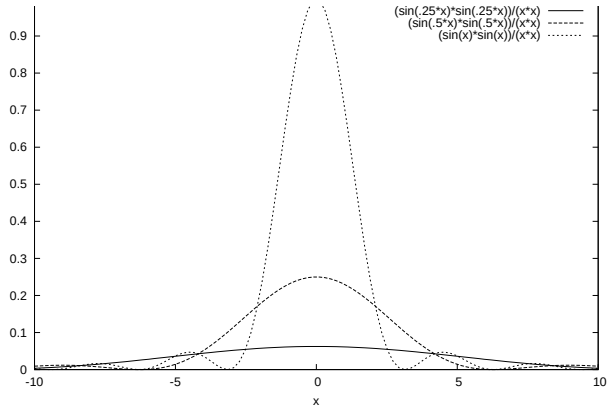


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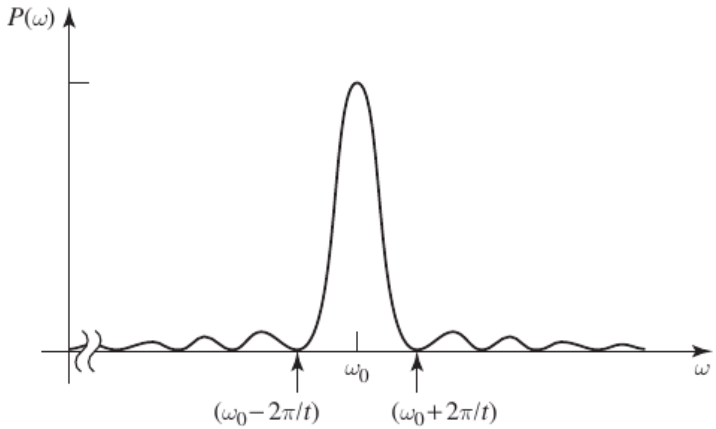


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$\therefore$ area of central peak goes as $2\pi t$



EMISSION AND ABSORPTION OF RADIATION

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Evidently $H'_{ba} = -\mathfrak{p} E_0 \cos(\omega t)$, where $\mathfrak{p} \equiv q\langle\psi_b|z|\psi_a\rangle$: transition dipole moment

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: stimulated emission
: Electromagnetic field gains energy from the system

Spontaneous emission : system in excited state makes a transition downward with the release of a photon, but without any applied electromagnetic field to initiate the process

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net transition probability $P_{b\to a}(t) = \frac{2}{\epsilon_0 \hbar^2} |\mathfrak{p}|^2 \int_0^\infty \rho(\omega) \left\{ \frac{\sin^2[(\omega_0 - \omega)t/2]}{(\omega_0 - \omega)^2} \right\} d\omega$

The energy density in an electromagnetic wave is $u = \frac{\epsilon_0}{2} E_0^2$ where $E_0 =$ amplitude of electric field

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Conclusion: transition rate for stimulated emission from state b to state a under influence of incoherent, unpolarized light incident from all directions,

$$R_{b\to a} = \frac{\pi}{3\epsilon_0 \hbar^2} |\mathfrak{p}|^2 \rho(\omega_0) \tag{1}$$

For systems like hydrogen, matrix elements obey selection rules
$\Delta l \equiv l' - l = \pm 1, \quad \Delta m \equiv m' - m = 0$ or ± 1
If these conditions are not met, the matrix element is zero, and the transition is said to be forbidden.
When E_f falls in a continuum of states, the transition rate is given by Fermi's Golden Rule :

$$R = \frac{2\pi}{\hbar} \left| \frac{V_{if}}{2} \right|^2 \rho(E_f). \tag{2}$$

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We can find out if these integrals are zero or not using symmetry considerations